Test Normal Cases

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Write expectations. Keep every result. Report the full set.

Compare a fixed activity prototype and baseline, inspect a supplied failure and explain the limits of routine coverage.

Keep one habit: Precommit the checks → record actual runs → count full passes without deleting failures.

MAIN LESSON · SECTION 1

What counts as a full pass?

> Fix the version and judging rules before comparing methods.

Start with a misleading success

Write the required result before running a test, then keep every observed result. A right label with a false explanation is not a full pass under this project's contract.

Recall expected versus observed results

How does an **expected result** differ from an observed result, and why keep both when they disagree?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Your learning goals

- I can plan routine cases with **expected result**s derived from the requirements.
- I can compare a fixed prototype and manual **baseline** on the same requests.
- I can report every result and explain what a small **test set** cannot establish.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

Repeated success on ordinary cases is necessary but not sufficient.
</details>



LEARNING TARGET A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can test routine cases against expectations and count full passes honestly.

Test the unchanged activity prototype fairly

> Use catalogue B version 1 and the same contract for P1 and the manual baseline.

Turn demonstrations into an inspectable test record

This week you will turn separate paper traces into a test record. Choose several ordinary requests, write the required answers first, then run the same saved prototype on every request.

Testing is useful when it can reveal a mistake, so keep failures as carefully as successes. A demonstration chosen because it works is not a complete evaluation.

Read the complete catalogue and requirements

Use catalogue B version 1 in order: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These fictional values assume materials are available. **Input**s are whole minutes from 0 to 30 and Quiet required Yes or No. Time must fit AND either quiet is not required or the activity is quiet. No allows either kind.

Select the first fit with a correct reason; otherwise report no match. Missing or invalid **input** requires clarification. No booking or publishing takes place.

Keep both methods and reset rules explicit

Keep the P1 version 1 instructions from Week 32: validate and stop with clarification if **input** is invalid; otherwise start at B1, check both conditions, suggest the first passing row and stop.

After a failure advance one row; report no match only after all rows fail. Reset to validation and B1 for each new valid request. The manual **baseline** marks every row against both conditions and selects the first fit.

Both methods must follow the same contract.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY The sorter is tested on five common classroom items with expected bins written before each run.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Prepare the four-case routine set before running it

> Choose cases for what they reveal and record both outputs and explanations.

Inspect the test-record structure

From a planned case to evidence

BEFORE THE RUN	DURING THE RUN	AFTER THE RUN
Write input and expected result	Use the saved version	Compare with the requirement
State why the case matters	Record output and reason	Mark pass or explain failure
Label both comparison methods	Keep every trace	Report counts and limits

Keep the catalogue, requests and pass criteria the same for the prototype and baseline.

Define labels, trace references and a full pass

Give each case an ID. Record its exact **input**, purpose, expected **output** and expected explanation before running it. Then add observed **output**, observed explanation, pass or fail, and a trace reference.

Put the prototype version, catalogue version and **baseline** method at the top of the sheet.

A pass needs the correct result and a reason supported by the checked conditions; the right activity with a false reason is not a full pass.

Read R1–R4 and each case's purpose

Our routine set R contains four valid requests. R1 is 11 minutes Yes, checking an early quiet match. R2 is 9 minutes No, checking that the search reaches a later **fitting** row.

R3 is 6 minutes No, checking that a valid request can exhaust the list. R4 is 9 minutes Yes, checking that a time fit alone is insufficient. These cases do not cover every accepted **input** or validation error.

Explain the test process

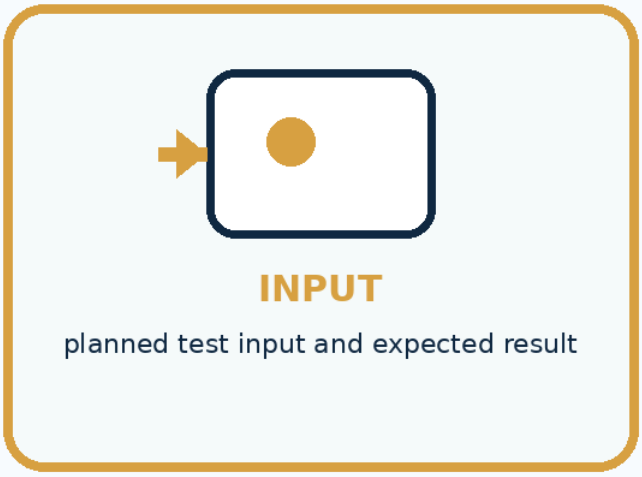
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

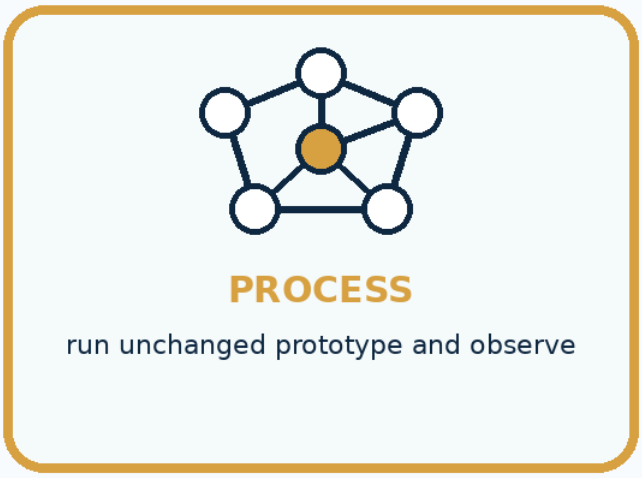
▼ Earlier wording, alternatives and picture transcripts

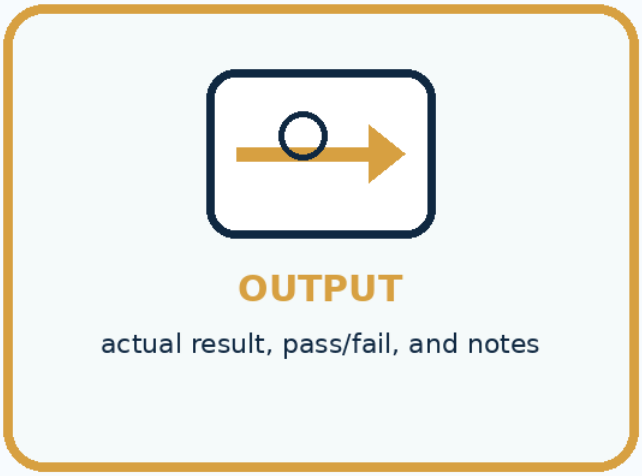
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT planned test **input** and **expected result**

PROCESS run unchanged prototype and observe

OUTPUT actual result, pass/fail, and notes

CORE IDEA Repeated success on ordinary cases is necessary but not sufficient.

UNDER THE HOOD A **test case** includes **input**, setup, expected behavior, actual behavior, and result. Precommitting expectations reduces hindsight bias.

Repeated normal tests establish basic reliability but say little about uncommon conditions.

Draw a fresh example with all three stages:

A closer explanation

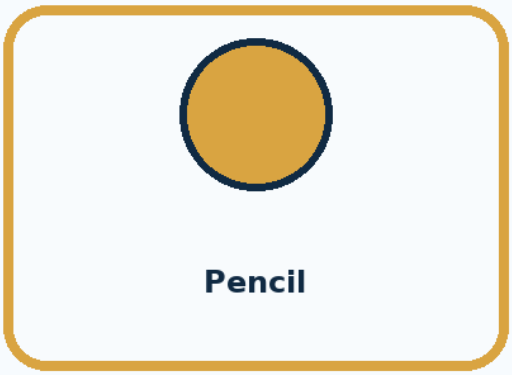
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare reproducible tests with selected demonstrations

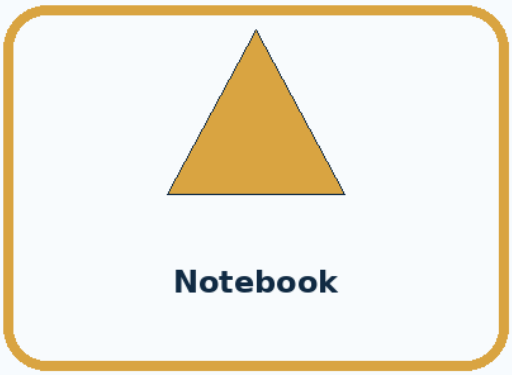
> Keep expectations, setup and failures visible.

Pictures paired with their evidence



The case to investigate

CASE 1 — Pencil Clue: common clear item. Meaning: normal case.



The case to investigate

CASE 2 — Notebook Clue: common clear item. Meaning: normal case.



The case to investigate

CASE 3 — Expected written later Clue: changed after result. Meaning: weak **evidence**.



The case to investigate

CASE 4 — Same **process** each run Clue: controlled setup. Meaning: **reproducible**.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

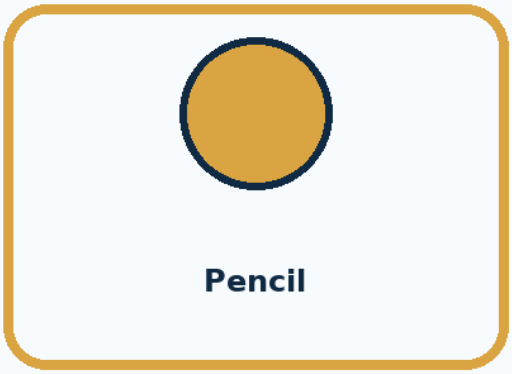
BOUNDARY Do not change the prototype halfway through a **test set** and combine results as if one version was tested.

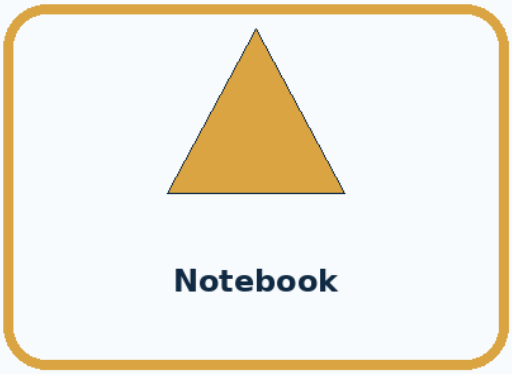
▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Pencil Clue: common clear item. Meaning: normal case.

CASE 2 — Notebook Clue: common clear item. Meaning: normal case.

CASE 3 — Expected written later Clue: changed after result. Meaning: weak **evidence**.

CASE 4 — Same **process** each run Clue: controlled setup. Meaning: **reproducible**.

Change one thing in the pictured case

Change: Prototype reports none for request 6 instead of P1.

Prediction: That run still fails.

Reason: P2 can hold 6, so none is also wrong under the first-fit rule.

Audit the separate pencil-pouch test record

> These P1/P2 labels identify pouches, not the activity prototype’s instruction versions.

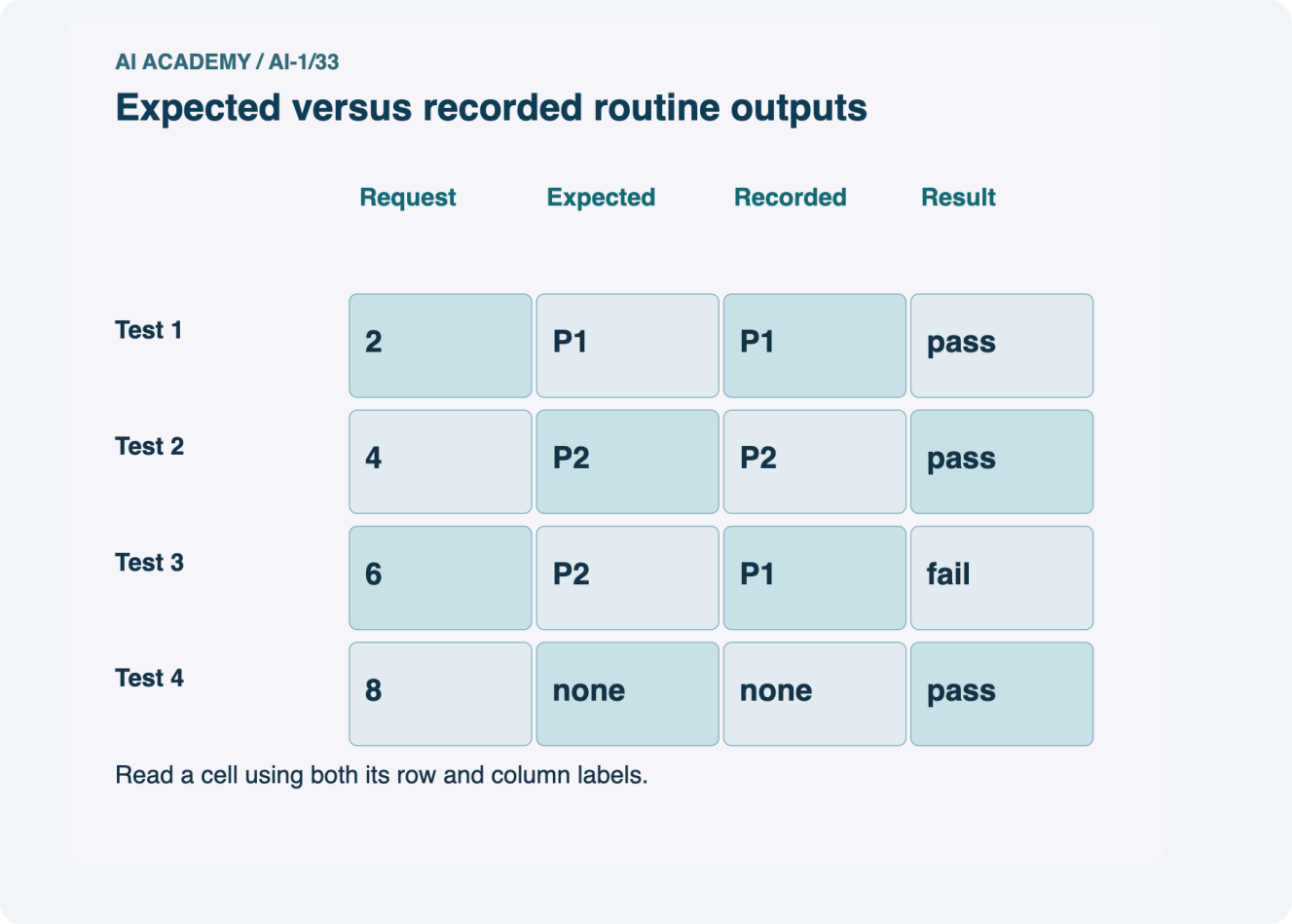
Read the pouch catalogue and recorded outputs

A fictional pencil-pouch finder has been run on four ordinary requests.

Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9.

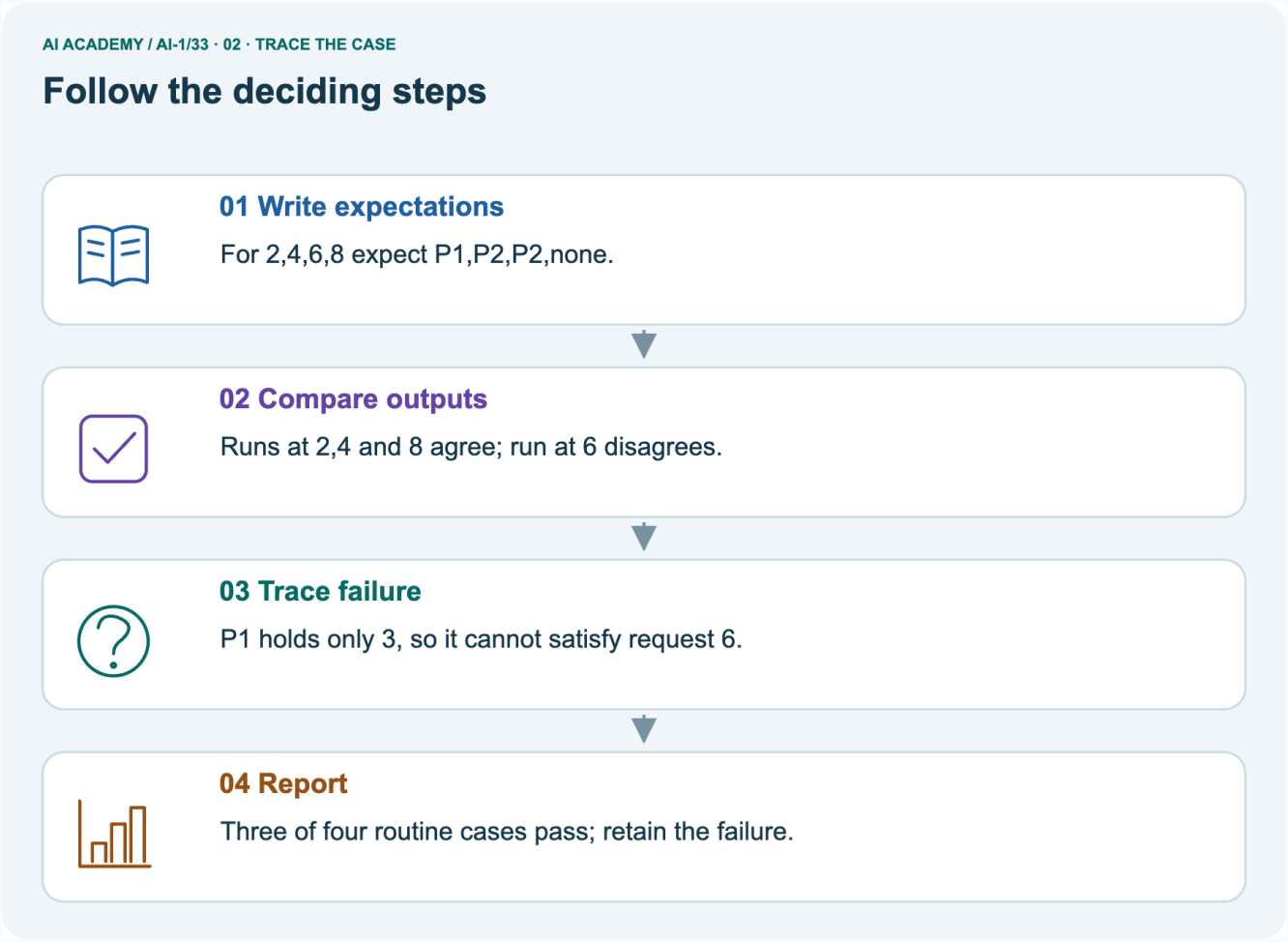
Requests 2,4,6,8; recorded **output**s P1,P2,P1,none. These are invented records, not class results.

Inspect the labelled routine-test comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace expectations, mismatch and full count



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Write expectations	For 2,4,6,8 expect P1,P2,P2,none.
2. Compare outputs	Runs at 2,4 and 8 agree; run at 6 disagrees.
3. Trace failure	P1 holds only 3, so it cannot satisfy request 6.
4. Report	Three of four routine cases pass; retain the failure.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain why no match depends on the request

- An ordinary case is not automatically correct. Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

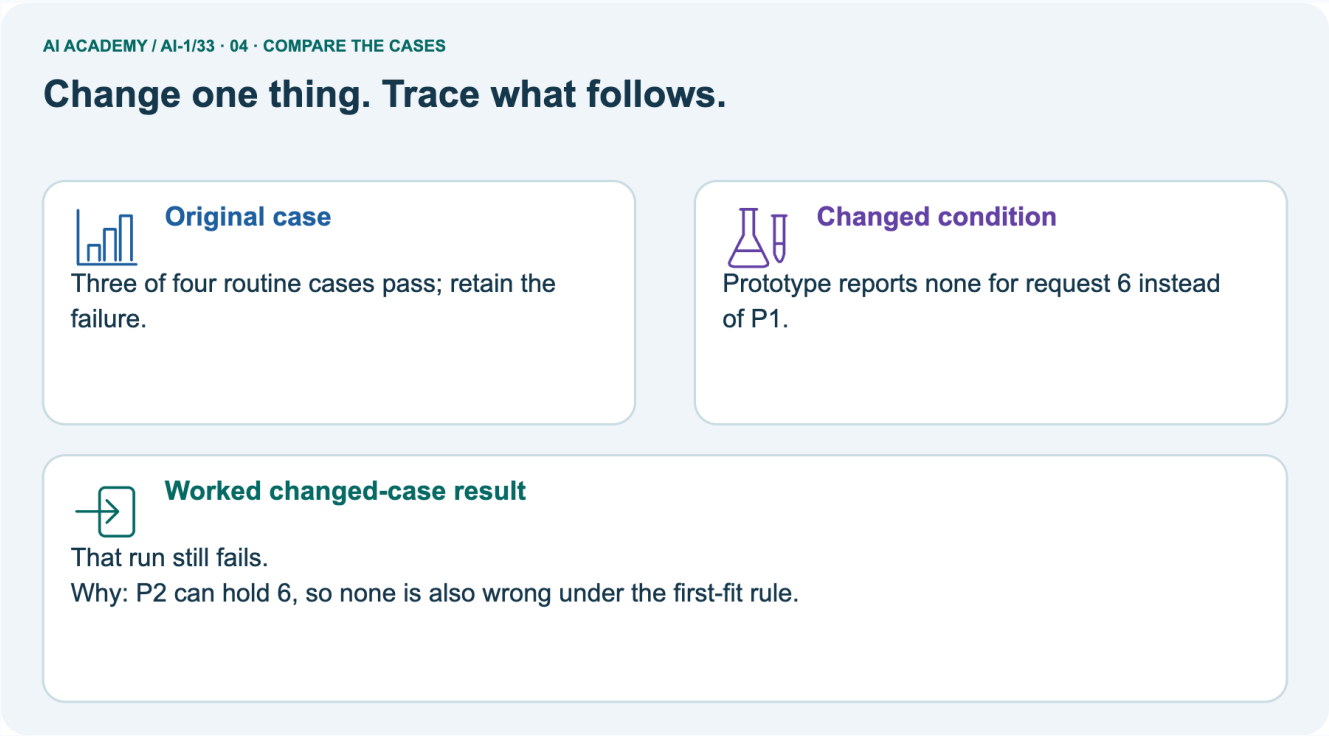
  A no-match **output** can be a full pass when that is exactly what the valid request requires.

- Read the labelled picture: Why is none correct for 8 but incorrect for 6? Use both capacity cards.

Predict an alternative wrong output

- Before opening the changed-pouch output, replace the wrong P1 suggestion for request 6 with no match. Does that repair the failure when P2 has capacity 7? State the required result before checking.

▼ Check the changed record



Follow the labelled picture with the matching explanation. Open the vector diagram

Name untested conditions

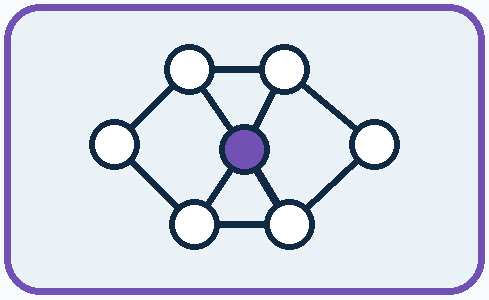
Four routine tests do not cover equality, invalid **input**s or every relevant classroom condition.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE The sorter is tested on five common classroom items with expected bins written before each run.

STEP 1 — OBSERVE planned test **input** and **expected result**

STEP 2 — TRACE run unchanged prototype and observe

STEP 3 — CONCLUDE **actual result**, pass/fail, and notes

MODEL REASONING Write the expected **output** first, keep the prototype version fixed, record every actual **output**, and count passes without deleting failures.

Explain that five normal successes are necessary **evidence**, not full proof.

DO NOT OVERCLAIM Do not change the prototype halfway through a **test set** and combine results as if one version was tested.

What new **evidence** could change this answer?

A closer explanation

Case and request	Expected output and reason	Fictional observed Q version 0
R1 11 minutes Yes	B1; 10 fits 11 and B1 is quiet	B1 with the same correct reason; pass
R2 9 minutes No	B2; 8 fits 9, quiet unrestricted, B1 too long	B2 with the same correct reason; pass
R3 6 minutes No	No match; all activities take longer	No match with the same correct reason; pass

Case and request	Expected output and reason	Fictional observed Q version 0
R4 9 minutes Yes	No match; B2 is not quiet and other rows take longer	B2 because 8 fits 9; fail

This is an invented record for a comparison prototype Q version 0, not a claim about your P1 runs. Q ignores the quiet condition when searching. Its trace explains the R4 failure.

In the same fictional comparison, the manual **baseline** produced the **expected result** and correct explanation on all four cases: 4 of 4 full passes. Q achieved 3 of 4. The denominator includes the failed case.

On this set Q has fewer full passes than the **baseline**. This supports investigating its quiet check; it does not establish performance on all requests. Do not claim a speed advantage because we did not measure time.

Correct implementations of P1 and the **baseline** can both pass all four, in which case these results show no correctness advantage for either method.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Test Normal Cases

The facts we will use

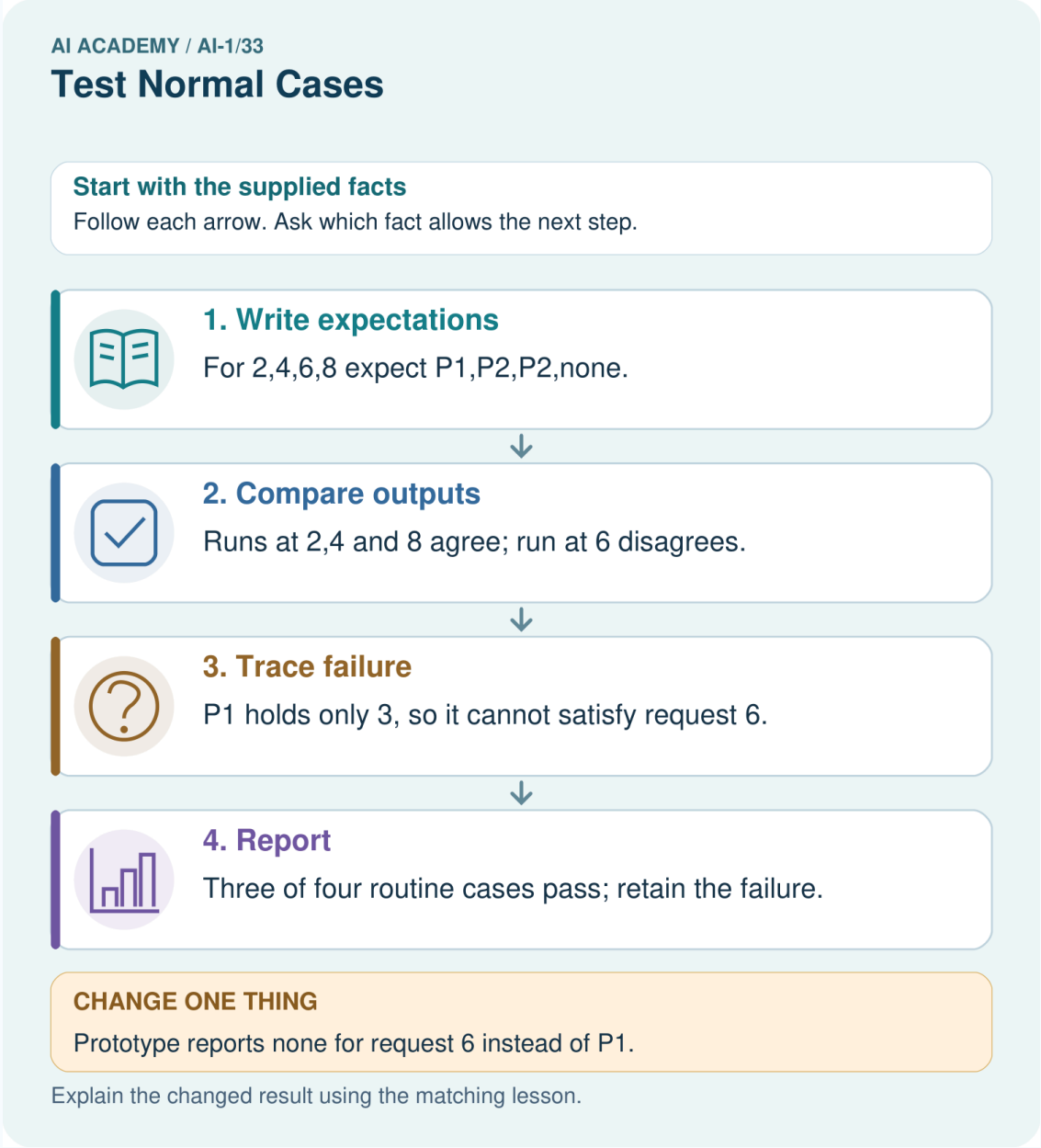
AI ACADEMY / AI-1/33

Expected versus recorded routine outputs

	Request	Expected	Recorded	Result
Test 1	2	P1	P1	pass
Test 2	4	P2	P2	pass
Test 3	6	P2	P1	fail
Test 4	8	none	none	pass

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: Prototype reports none for request 6 instead of P1. Predict the result before checking the explanation. What remains the same?

Report counts, failures and coverage honestly

> Keep a result about this set separate from a claim about all inputs.

Write a reproducible limited report

A useful report names the version, catalogue, case set, full-pass count and total, then describes each failure and the requirement it breaks. Keep the trace so someone can reproduce the result.

Say which conditions have not been tested. Four passing runs support a claim about those four runs; they do not prove that every possible request works.

Explain why B3/B4 cannot be first outputs in this catalogue

For catalogue B, any request allowing B3 also allows the earlier quiet B1; any request allowing B4 also allows earlier B1 when quiet is not required. Therefore the first-match rule never selects B3 or B4 with this catalogue.

Checking those rows in a no-match trace is different from testing a successful **output** from a later row.

Next week we will deliberately examine boundaries and labelled catalogue variations while keeping the requirements explicit.

Repair both tempting claims

- CLAIM A Five passes prove the system is ready for every item.
- CLAIM B A failed test should be removed because it lowers the score.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Pencil	common clear item	normal case
Notebook	common clear item	normal case
Expected written later	changed after result	weak evidence
Same process each run	controlled setup	reproducible

Repair two tempting claims

Repair:

Repair:

A closer explanation

> **A team reports "Our prototype passed 4 of 4 routine cases, so it always works and is faster than the baseline." Rewrite this as a claim supported by the evidence, and identify two further checks that would address different gaps.**

MAIN LESSON · SECTION 8

Practise: run both methods without choosing only successes

> Preserve one fixed-version set before starting a separately labelled repaired set.

Precommit expectations and record actual operation

Before testing your own cards, finish all four **expected result**s and agree the **pass criterion**. Swap roles: one pupil operates and the other records.

Keep the same version throughout the set and begin a fresh run for each request. Record what happened even if a card is ambiguous or the operator makes a mistake.

A run that stops without a required answer is a failed run to investigate, not a row to delete.

Run the same baseline cases and retain all versions

Run the manual **baseline** on the same four requests and catalogue. Record its **actual result**s too; a person can make an error in either method.

If you find a defect, preserve this set's results, save a new version and start a separately labelled set after the repair. Never combine the best result from each version into a score for one version.

Run the fair comparison in W2

Use workbook W2 for the connected example "Testing routine cases". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge the label-alone pass claim

- Use the worked case for Testing routine cases. Proposed explanation: "A correct label is always a full pass". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the pouch no-match contrast

Why is none correct for 8 but incorrect for 6? Use both capacity cards.

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Repeated success on ordinary cases is necessary but not sufficient.

Use a fresh example to explain **test case**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: A failed test should be removed because it lowers the score.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

> **In the fictional Q record, give the full-pass fraction for each method and explain the R4 failure. A teammate removes R4 and reports Q as 3 of 3. Explain how that changes the question being answered. Then state one condition that set R has not checked.**

Practise with less help: Run a fair comparison

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Investigate a complete five-case comparison

> Use the same cases, catalogue and judging rule for both methods.

The development cases and preparation

Five-case paper test plan for your saved P1 activity cards and the manual baseline. Use catalogue B version 1 and cases T1=(11,Yes), T2=(9,No), T3=(6,No), T4=(9,Yes), T5=(12,No). These are development cases, not the teacher's reserved final set. Write every required

output, explanation and case purpose before operating either method.

Run, retain and report every result

1. Record the actual result of each method on the same five requests, including errors, pauses and incomplete runs. Use one saved version per set and a full pass only for the correct result with a supported explanation. Report full passes out of all five attempted cases separately for each method; retain failures. A repair starts a newly labelled set, not an overwritten row. Do not infer speed or usability because neither is measured. If you have not operated the cards, mark the observed columns “not run”.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep traces, versions and limits

Case	Prediction	Observed result	What it means
Normal			
Edge			

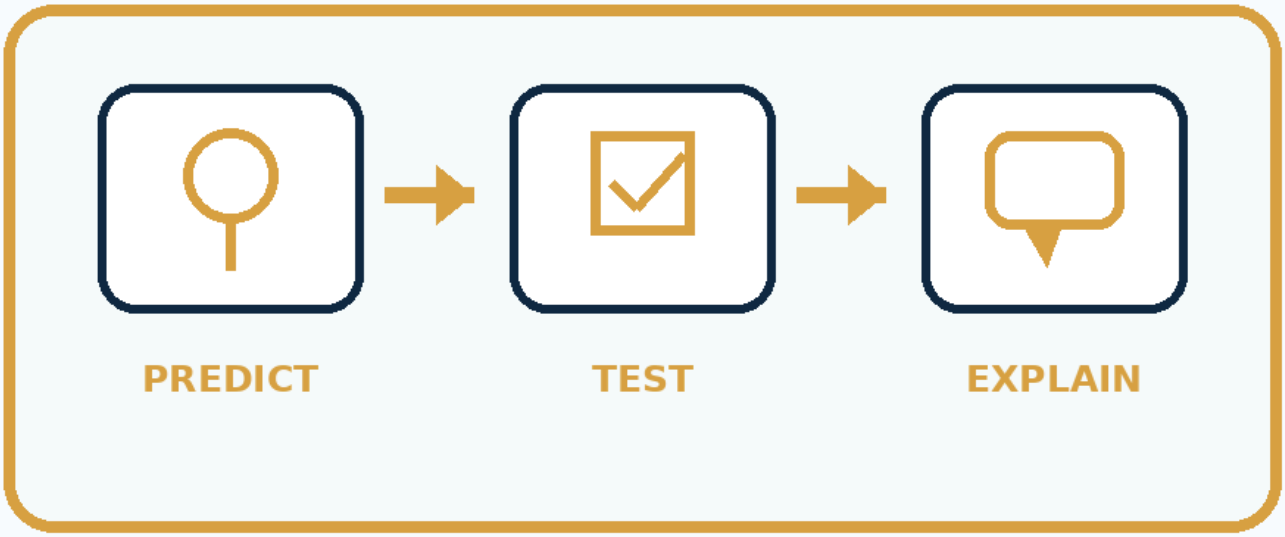
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Five-Case Normal Test

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Run five planned tests and log expected versus actual.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the new equipment test record

> Derive expectations before judging the three supplied outputs.

The independent catalogue and saved results

Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10.

Select the first item with capacity at least the group size, or no match; invalid **input** needs clarification.

The saved prototype reports E1 for group 3 with reason “4 holds 3”; E1 for group 6 with reason “4 holds 6”; no match for group 9 with reason “both too small”.

Write all **expected result**s, mark full passes, calculate the fraction and explain the failure. Give one untested **input** category. No booking occurs.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit a new test record

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Week 33 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Test a sorter on two clear pencil cards and two clear book cards. It gets one book wrong. Report the result and keep the failed case.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and **expected result**

> **Write or draw your reasoning and evidence.**

Connect to your project

Create and run the sorter's normal-case test plan with unique IDs and version label.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Transfer the testing method to the sorter portfolio

> Retain a complete result set for each context.

Connect the two contexts

The main sequence tests the activity prototype. Apply the same expected/observed/full-pass method to existing sorter evidence here, without requiring a second build. Keep each project's case IDs, catalogue and results separate; an activity pass count is not a sorter pass count.

This week's sorter test record

THIS WEEK'S CHECKPOINT Create and run the sorter's normal-case test plan with unique IDs and version label.

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment.

Inspect tests and necessary evidence

- What real classroom need does this serve?

- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Keep versions separate

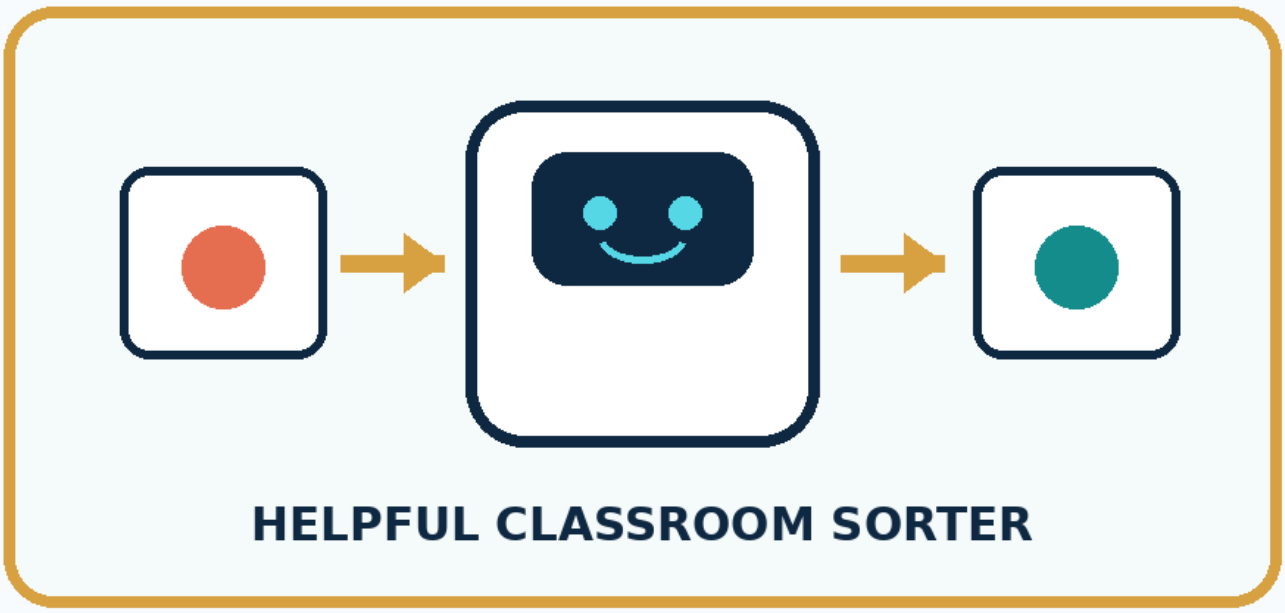
RESPONSIBLE DESIGN Do not change the prototype halfway through a **test set** and combine results as if one version was tested.

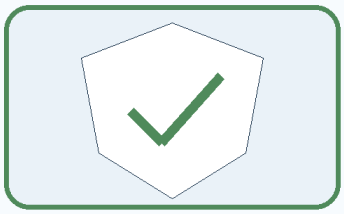
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: report a complete result and a coverage limit

> Close your notes and explain why a retained failure is useful evidence.

Your exit explanation

- Close your notes. Explain this goal in your own words: Plan routine cases with **expected result**s derived from the requirements

My deciding step, evidence and one remaining question

Choose the next revealing test

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
test case	planned input and expected behavior	
expected result	prediction written before running	
actual result	observed behavior	
reproducible	another tester can repeat the setup	

Terms for the connected explanation

Word	Meaning
Test set	A collection of cases used together to check a version
Pass criterion	The requirement an observed result must meet to count as a pass
Coverage	Which conditions or behaviours the selected tests examine

Word	Meaning
Baseline	A simple comparison method tested under the same conditions

Repair a claim using a word

- CLAIM TO REPAIR Five passes prove the system is ready for every item.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

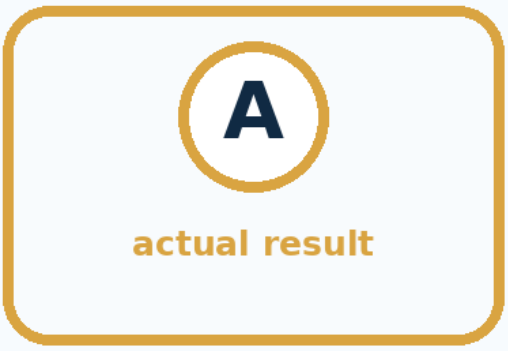
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

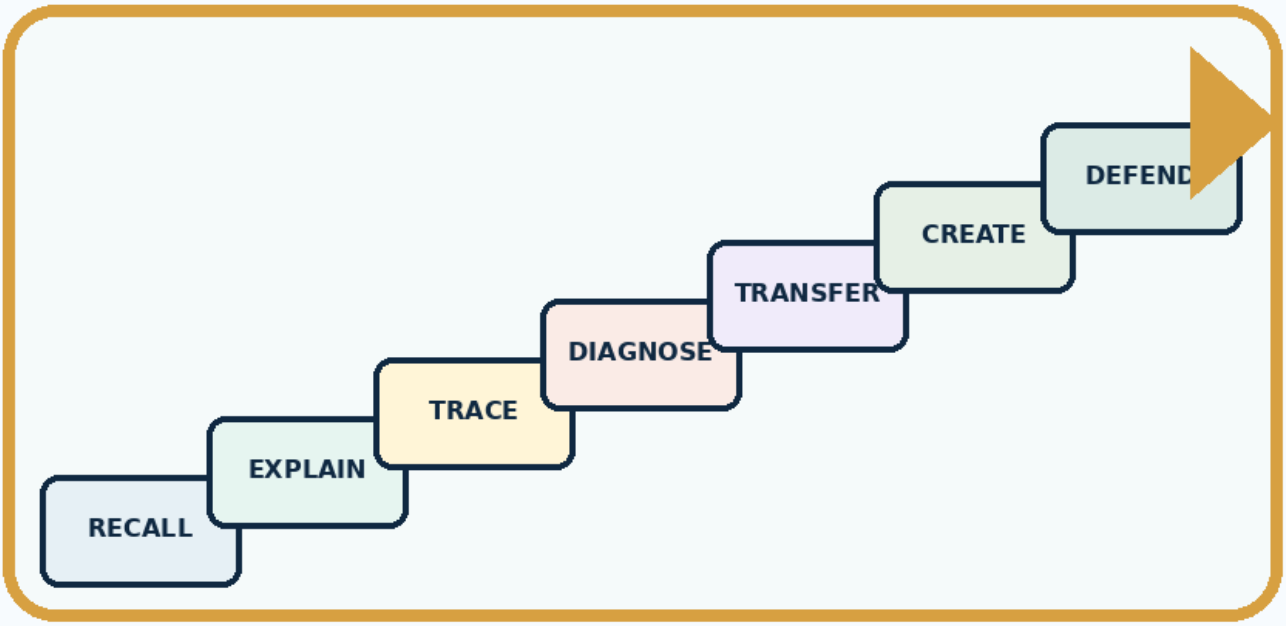
- RECALL Define **test case**.
- EXPLAIN Repeated success on ordinary cases is necessary but not sufficient.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Five passes prove the system is ready for every item.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Audit a new test record is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

Checklist for explaining this week's idea

Use after practice. Audit a new test record is the selected independent case.



EXPLANATION CHECK Derive expectations from the contract, judge outputs and reasons, keep every case in the denominator and limit conclusions to tested conditions.

Independent **evidence**:

SELF-CHECK Write the expected **output** first, keep the prototype version fixed, record every actual **output**, and count passes without deleting failures. Explain that five normal successes are necessary **evidence**, not full proof.

NEXT BRIDGE Carry today's **evidence** habit into: Test Edge Cases and Fairness.

▼ **Extra practice: Report the cases actually tested**

REFERENCE · SECTION 13

Extra practice: Report the cases actually tested

> Try the question before opening its worked explanation.

Read the supplied case and question

Four routine cases are chosen before testing. Three are correct.

What can be reported?

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

3/4 on those cases. Preserve the failure and trace its cause. This does not guarantee unfamiliar-case performance.

Evidence or stage	Value or result
Routine cases	3/4

▼ Earlier wording, alternatives and picture transcripts

Week 33 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Report the cases actually tested

Four routine cases are chosen before testing. Three are correct.

What can be reported?

3/4 on those cases. Preserve the failure and trace its cause. This does not guarantee unfamiliar-case performance.

▼ Reference: original lesson checklist

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Plan routine cases with expected results derived from the requirements	Explain the deciding step, show a trace or test, and state one limit.
Compare a fixed prototype and manual baseline on the same requests	Explain the deciding step, show a trace or test, and state one limit.
Report every result and explain what a small test set cannot establish	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional pencil-pouch finder has been run on four ordinary requests.

▼ Extra practice: pictures, repair and feedback

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional pencil-pouch finder has been run on four ordinary requests.
- Which labelled step matches this detail? Three of four routine cases pass; retain the failure.
- 1. Trace failure
- 2. Write expectations
- 3. Compare **output**s
- 4. Report
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report Three of four routine cases pass; retain the failure. An ordinary case is not automatically correct.

Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires.

Changed-case question

- Changed-case practice: Prototype reports none for request 6 instead of P1. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. That run still fails.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed case**

After your attempt, compare with the supported explanation: That run still fails. P2 can hold 6, so none is also wrong under the first-fit rule.

Repair and save your work

Review Build a Paper Prototype (ai-1/32). Goal: Build a runnable paper prototype with explicit instructions and visible state. Short practice: What should appear in the test log before and after the repair?.

Return to this week and explain what you changed.

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?.

Return to this week and explain what you changed.

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

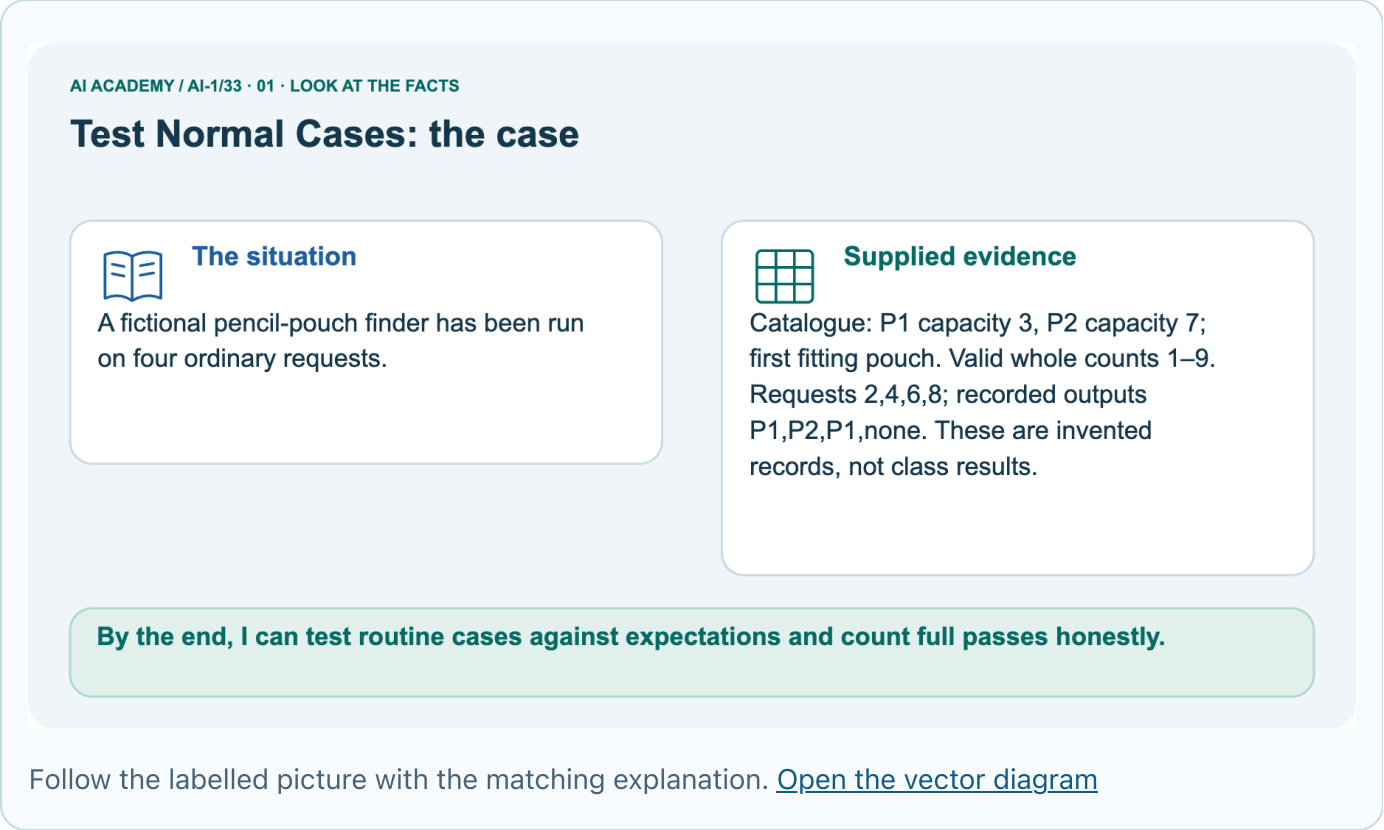
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Test Normal Cases: the case



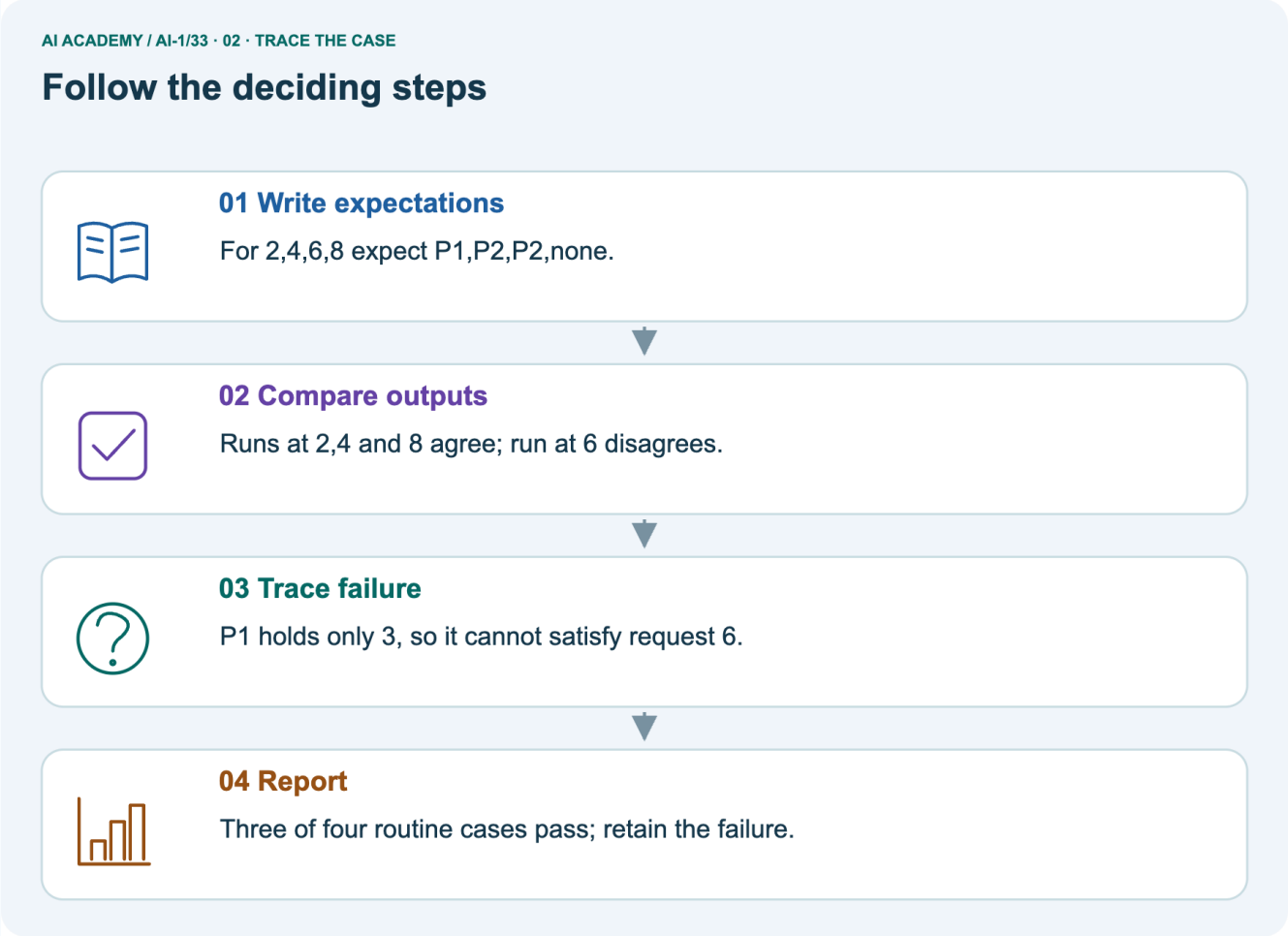
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional pencil-pouch finder has been run on four ordinary requests. Supplied facts: Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9. Requests 2,4,6,8; recorded **output**s P1,P2,P1,none.

These are invented records, not class results.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

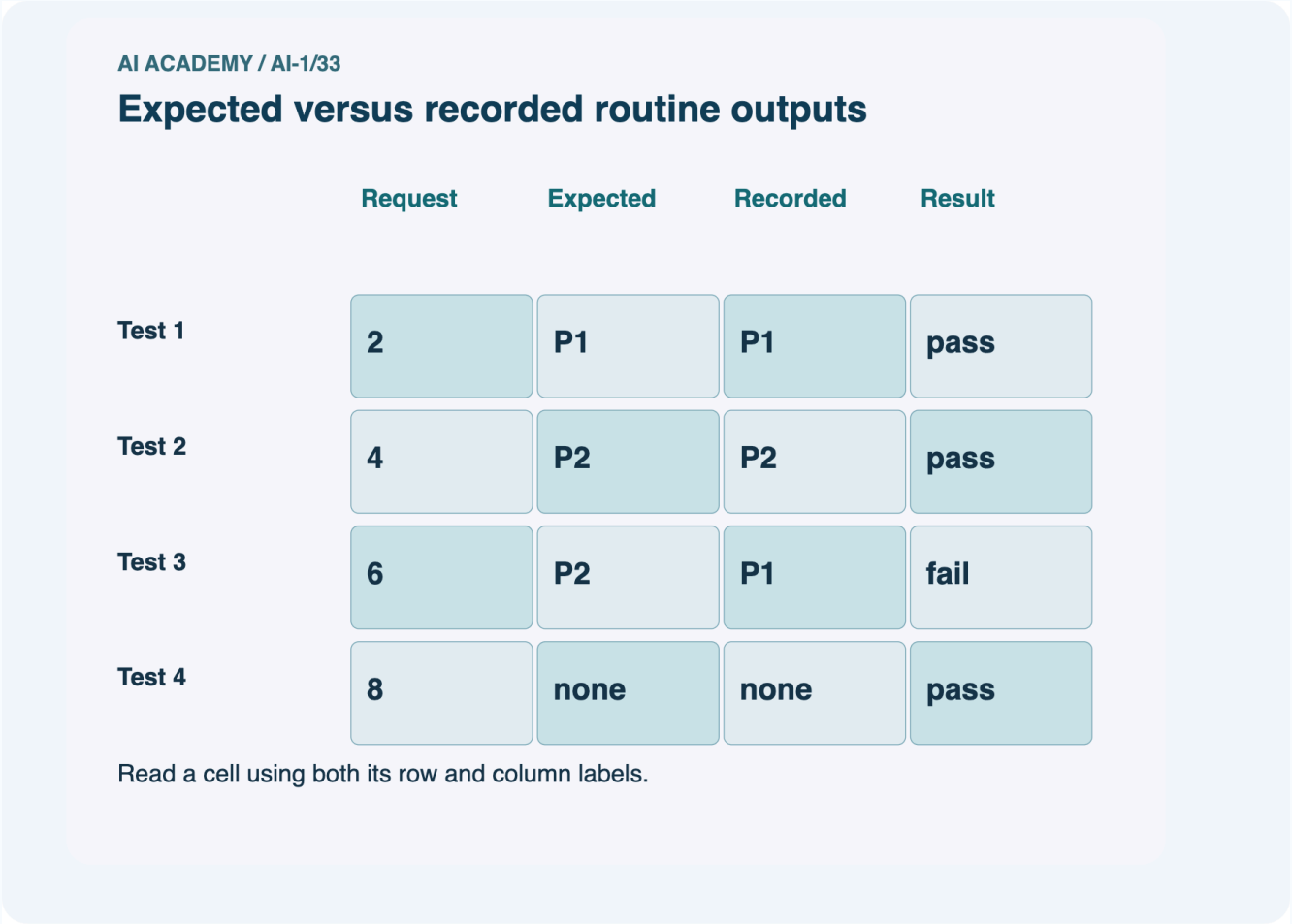


Follow the labelled picture with the matching explanation. Open the vector diagram

Write expectations: For 2,4,6,8 expect P1,P2,P2,none. → Compare **output**s: Runs at 2,4 and 8 agree; run at 6 disagrees. → Trace failure: P1 holds only 3, so it cannot satisfy request 6. → Report: Three of four routine cases pass; retain the failure.

Point to each numbered stage and explain which recorded **input** it uses.

03. Expected versus recorded routine outputs



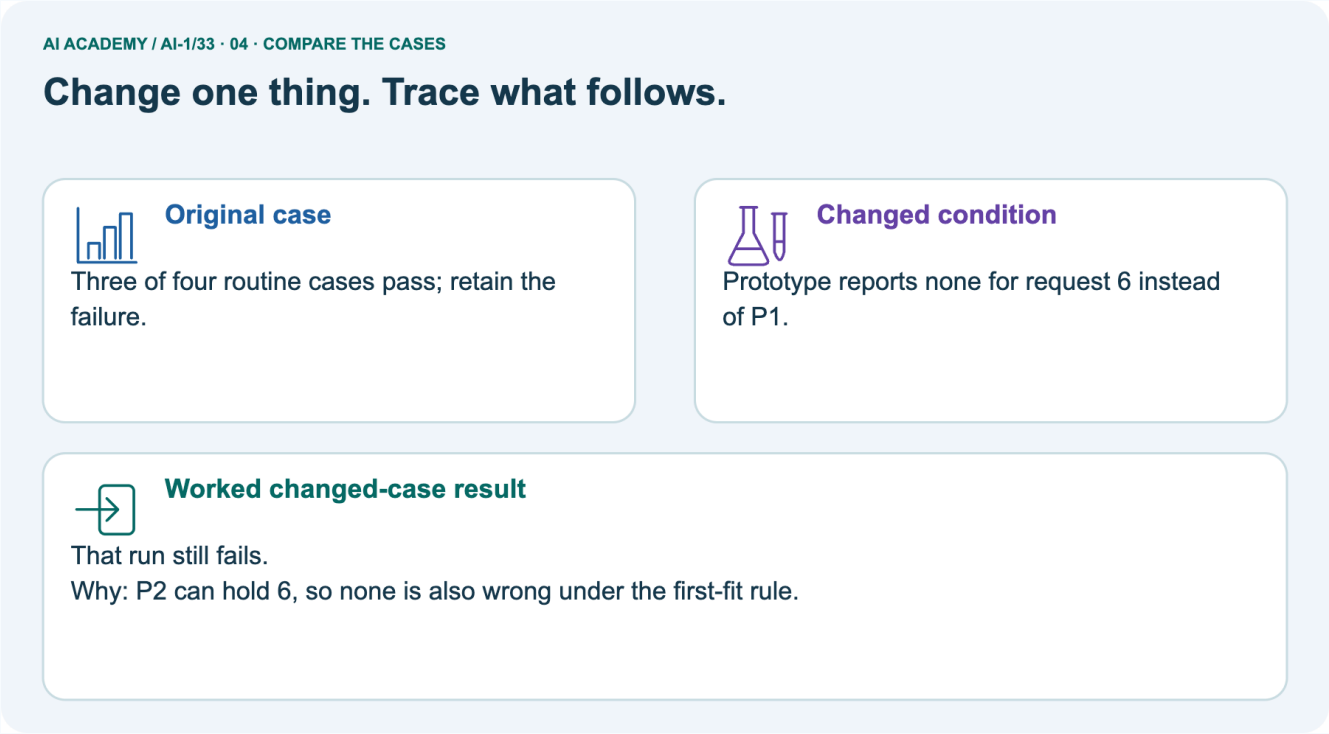
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected versus recorded routine **output**s. Test 1: Request = 2, Expected = P1, Recorded = P1, Result = pass. Test 2: Request = 4, Expected = P2, Recorded = P2, Result = pass.

Test 3: Request = 6, Expected = P2, Recorded = P1, Result = fail. Test 4: Request = 8, Expected = none, Recorded = none, Result = pass.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



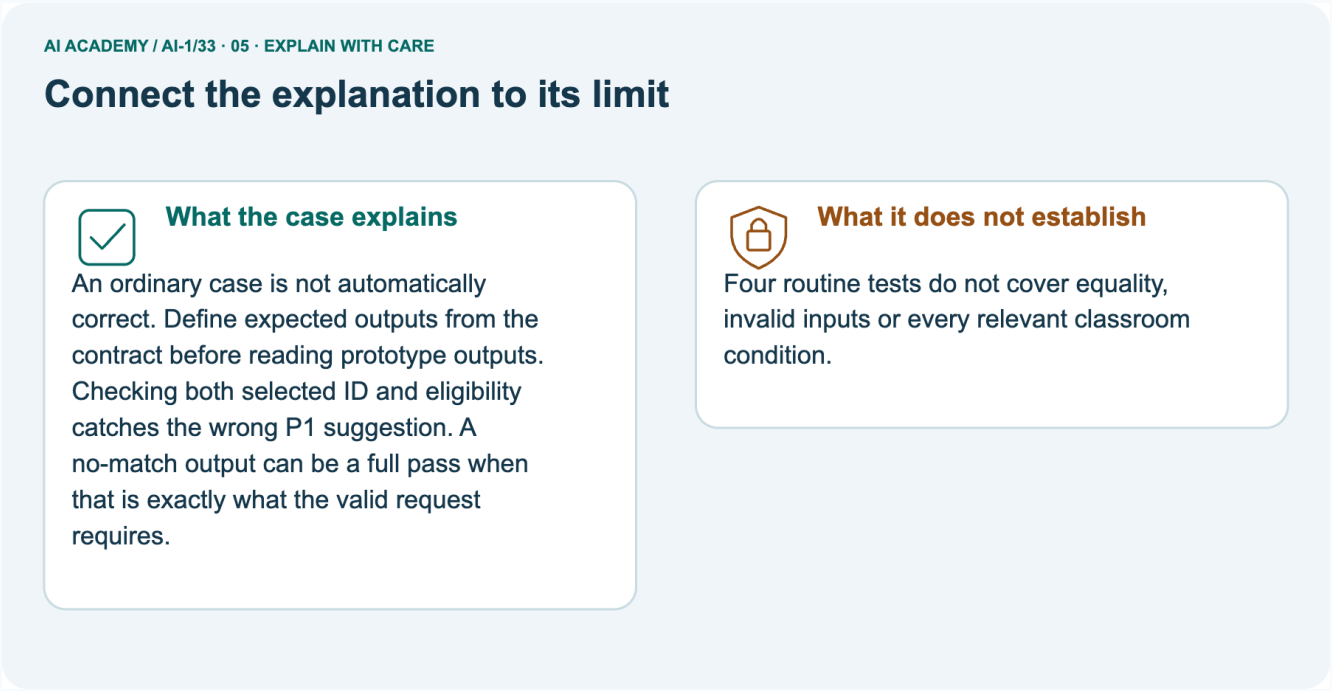
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three of four routine cases pass; retain the failure. Changed condition: Prototype reports none for request 6 instead of P1. **Expected result**: That run still fails.

Reason: P2 can hold 6, so none is also wrong under the first-fit rule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



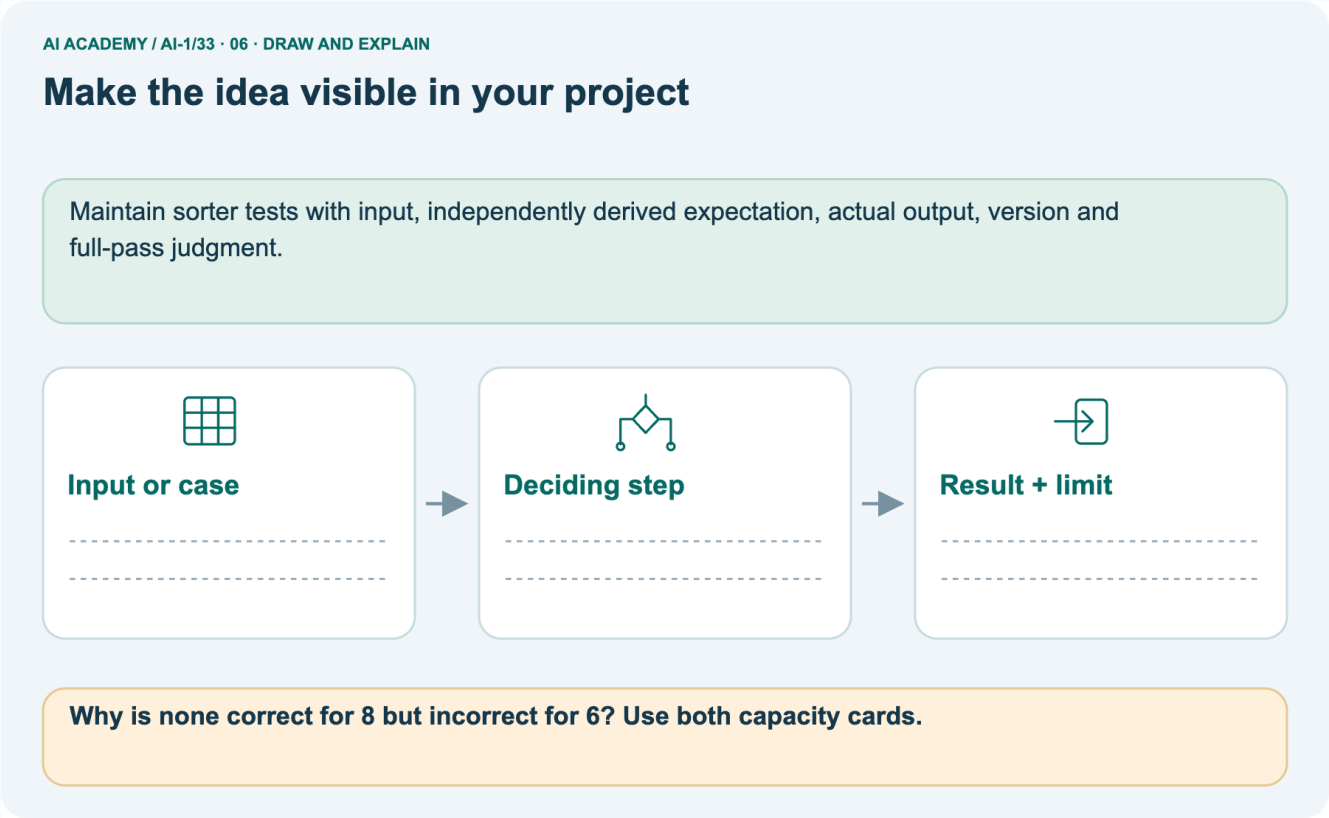
Follow the labelled picture with the matching explanation. Open the vector diagram

An ordinary case is not automatically correct. Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires. Limit: Four routine tests do not cover equality, invalid **input**s or every relevant classroom condition.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment. Draw your **input**, deciding step, result and limit.

Supported question: Why is none correct for 8 but incorrect for 6? Use both capacity cards.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Test Normal Cases: the case

02. Follow the deciding steps

03. Expected versus recorded routine outputs

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.